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Designing a new physical activity calorie equivalent food label and comparing its effect on caloric choices to that of the traffic light label among mothers: a mixed-method study

Shirin Seyedhamzeh, Saharnaz Nedjat, Elham Shakibazadeh, Azam Doustmohammadian, Hedayat Hosseini, Asma Kazemi, Nafiseh Azizolahi, Maryam Chamary, Cain C T Clark, Ahmadreza Dorosty Motlagh

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Abstract

We designed a new type of 'physical activity calorie equivalent' (PACE) food label in Iran to compare its effect with that of the traffic light food label (TLL) on caloric choices. Mixed-method study. Mothers of school children between the ages of 6-12 years. In the qualitative phase, 10 focus group discussions (FGDs) were conducted with various groups of mothers, and two FGDs were conducted with food science and nutrition experts to design a new PACE label. In the quantitative phase, 496 mothers were randomly assigned to five groups: (1) no nutrition label, (2) current TLL, (3) current TLL + educational brochure, (4) PACE label, and (5) PACE label + brochure. Samples of dairy products, beverages, cakes, and biscuits were presented. ANOVA and multiple linear regressions were applied to examine the association between label types and calories of the selected products as our main outcome. The mothers' perspectives were classified into two sub-themes, the PACE label's facilitators and barriers. The new PACE label's characteristics were divided into two subcategories: (a) appearance, and (b) nutritional information, including 14 codes. In the quantitative section, mean calories of the selected foods were lowest in the TLL + brochure group (831.77 kcal; 95% CI: 794.23-869.32), and highest in the PACE label group (971.61; 95% CI: 926.37-1016.84). The new PACE label was a combination of PACE, TLL, and warning labels. It did not significantly affect lower caloric choice, however, the TLL + brochure option was effective in choosing foods with fewer calories. : The study was registered in the Iranian Registry of Clinical Trials 23 (IRCT20181002041201N1). [Abstract copyright: Copyright © 2023 Seyedhamzeh, Nedjat, Shakibazadeh, Doustmohammadian, Hosseini, Kazemi, Azizolahi, Chamary, Clark and Motlagh.]

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Keywords

- nutrition label
- physical activity
- TLL
- obesity
- PACE

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KW - nutrition label

KW - physical activity

KW - TLL

KW - obesity

KW - PACE

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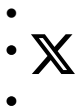
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